

Who Are Colluding?

A Linear-Algebraic and Statistical Audit of Test-Taking Behaviour SIMC 2026 Sample Problem – Tasks 1 through 5

May 16, 2026

Abstract

We treat the question “*which students colluded during a test?*” as a geometry problem on the design matrix $\mathbf{X} \in \{-1, +1\}^{M \times N}$. Tasks 1 and 2 visualise $\mathbf{X}$ and the Gram matrix $\mathbf{S} = \mathbf{X}\mathbf{X}^\top$, immediately exposing one pair of students (indices 5 and 10) whose answer vectors are *identical*. Task 3 applies K -means to recover the underlying group structure, and we introduce a **singleton-filtered silhouette criterion** that resolves the well-known tendency of the silhouette score to reward over-fragmentation. Task 4 makes the qualitative case for collusion. Task 5 closes the section by deriving the closed-form mean $\langle S \rangle = N(2p - 1)^2$ and standard deviation $\sigma_N = \sqrt{N[1 - (2p - 1)^4]}$ of the similarity score under a null hypothesis of independent test-takers; these will anchor the hypothesis tests of Tasks 6–8.

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1 Task 1 – The Design Matrix

1.1 Definition and encoding

Let there be M students and N questions. The *design matrix* is

$$\mathbf{X} \equiv X_{ij} \in \{-1, +1\}^{M \times N}, \quad X_{ij} = \begin{cases} +1, & \text{student } i \text{ answered question } j \text{ correctly,} \\ -1, & \text{otherwise.} \end{cases} \quad (1)$$

The symmetric ± 1 encoding (rather than $\{0, 1\}$) is deliberate: it makes Eq. (1) a sign-vector representation, so that the inner product of two rows directly measures *net agreement* (see Task 2).

1.2 Visualisation

Figure 1 shows the small subset `sample_small` $\in \{-1, +1\}^{50 \times 25}$ rendered as a binary heatmap, with marginal bar plots of the per-student total $\sum_j X_{ij}$ and per-question total $\sum_i X_{ij}$.

What the marginals already tell us

The per-question bars expose a markedly uneven test: questions 0, 7 and 23 behave like “free” questions (net positive $> +35$), while questions 19–22 are net negative. *Question difficulty is not uniform.* Task 8 will exploit exactly this fact by estimating per-question p_j from column means before applying the null model.

2 Task 2 – The Similarity Matrix

2.1 Construction

Define the *pairwise similarity matrix* $\mathbf{S} \in \mathbb{Z}^{M \times M}$ by

$$S_{ij} \equiv (\mathbf{X} \mathbf{X}^\top)_{ij} = \sum_{k=1}^N X_{ik} X_{jk}. \quad (2)$$

Because each factor is ± 1 , the contribution of question k to S_{ij} is

$$X_{ik} X_{jk} = \begin{cases} +1, & \text{students } i \text{ and } j \text{ **agreed** on question } k, \\ -1, & \text{students } i \text{ and } j \text{ **disagreed** on question } k. \end{cases} \quad (3)$$

Consequently

$$S_{ij} = \# \text{agreements} - \# \text{disagreements} \in \{-N, -N + 2, \dots, N\}, \quad (4)$$

with the diagonal pinned at $S_{ii} = N$ by construction (a student always agrees with themselves on every question).

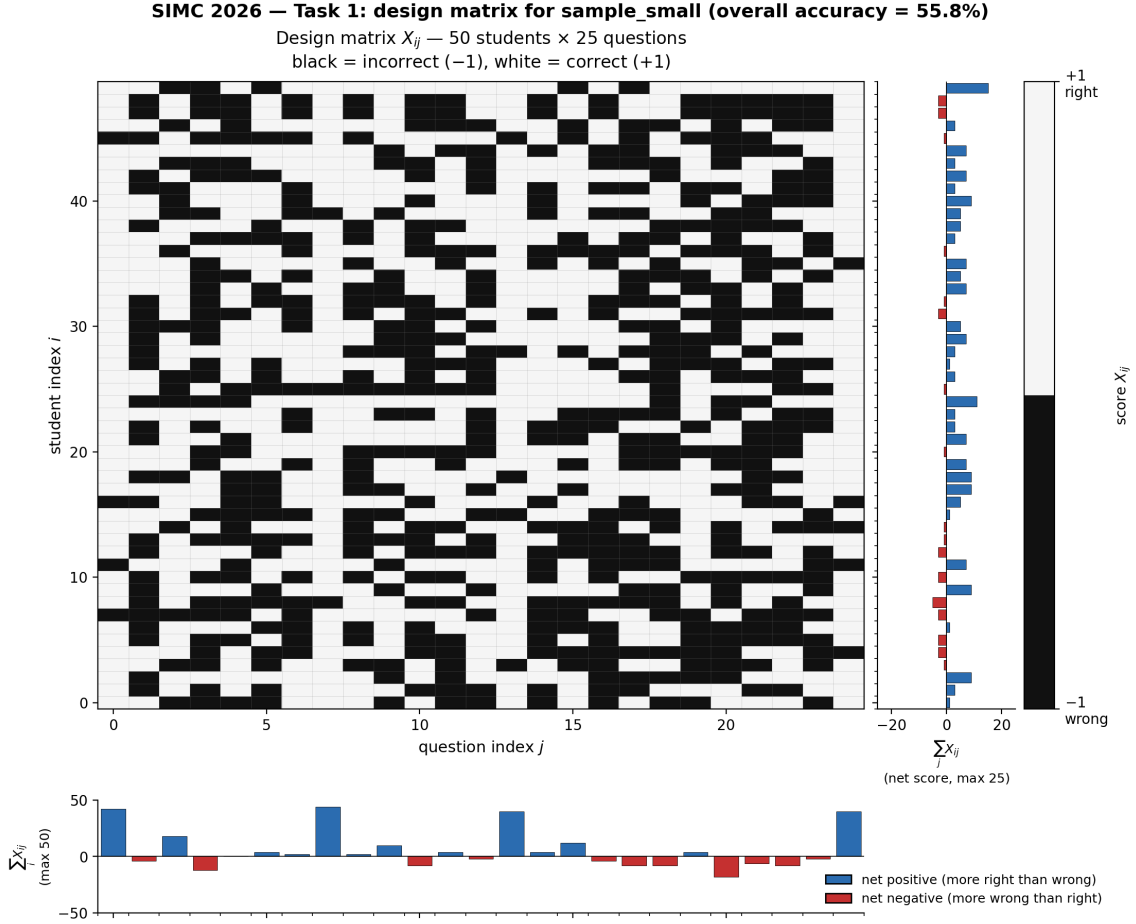


Figure 1: Design matrix for `sample_small` ($M = 50$, $N = 25$). Black cells encode wrong answers, white cells encode correct answers. Right margin: student net scores. Bottom margin: question net scores. Overall accuracy is 55.8%.

2.2 Geometric interpretation

Each row $\mathbf{X}_{i,\cdot}$ has constant Euclidean norm $\|\mathbf{X}_{i,\cdot}\| = \sqrt{N}$, so $\mathbf{S}$ is, up to a constant, a *cosine similarity* matrix:

$$\cos \theta_{ij} = \frac{\langle \mathbf{X}_{i,\cdot}, \mathbf{X}_{j,\cdot} \rangle}{\|\mathbf{X}_{i,\cdot}\| \|\mathbf{X}_{j,\cdot}\|} = \frac{S_{ij}}{N}. \quad (5)$$

This is the key bridge to clustering algorithms (Task 3): *any* distance-based clustering on the rows of $\mathbf{X}$ implicitly partitions students by cosine similarity, i.e. by $\mathbf{S}$.

2.3 Visualisation and outliers

Figure 2 plots $\mathbf{S}$ as a diverging heatmap together with the histogram of all $\binom{M}{2} = 1225$ unique off-diagonal entries.

A single pair sits at the theoretical maximum

The maximum off-diagonal similarity is $S_{5,10} = 25 = N$, meaning students 5 and 10 **agreed on all 25 questions**. The bulk of the histogram has mean $\bar{S}_{\text{off}} \approx 2.89$ with a roughly Gaussian spread, so the (5,10) entry is a striking outlier in the right tail.

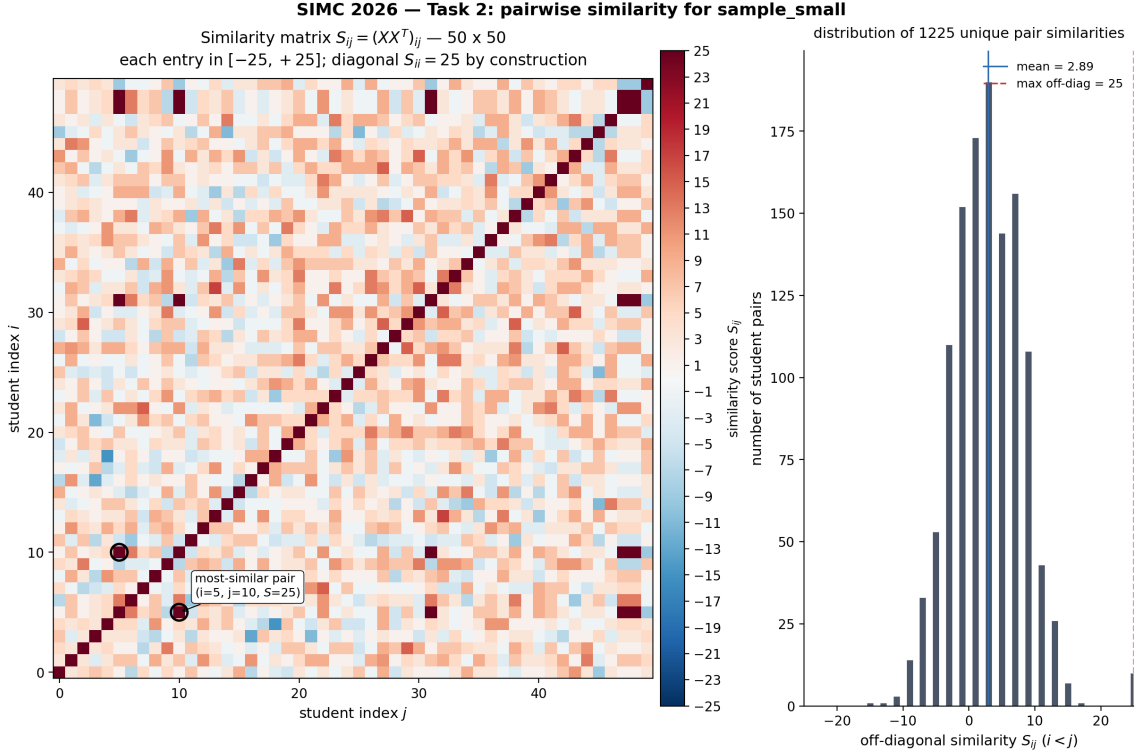


Figure 2: Left: similarity matrix $\mathbf{S} = \mathbf{X}\mathbf{X}^\top$ for `sample_small`. The diagonal $S_{ii} = N = 25$ is dark red. Right: histogram of off-diagonal S_{ij} with $i < j$.

3 Task 3 – Clustering with K -means and Silhouette Analysis

3.1 Algorithm

K -means partitions $\{\mathbf{X}_{i,\cdot}\}_{i=1}^M$ into K disjoint groups by alternately

- (a) assigning each student to the cluster whose centroid is closest, and
- (b) recomputing centroids as the mean of their assigned members,

until centroids stabilise. To mitigate sensitivity to the random initial centroids we use scikit-learn’s `n_init=20`.

3.2 Visualising cluster quality

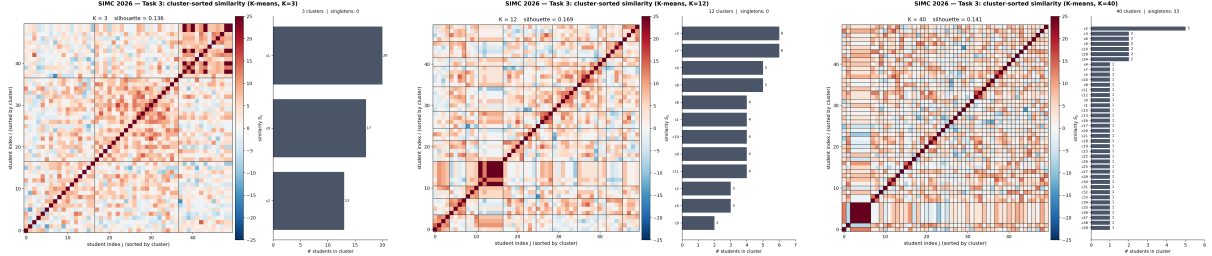
For each K we render $\mathbf{S}$ with its rows and columns permuted by cluster label. A good K produces solid red squares on the diagonal (in-group agreement) against a paler background (out-group disagreement). Figure 3 shows three representative values.

3.3 Silhouette score

For a single point i assigned to cluster $C(i)$, let $a(i)$ be its mean (cosine) distance to other in-cluster points and $b(i) = \min_{C \neq C(i)} \bar{d}(i, C)$ its mean distance to the nearest *foreign* cluster. The silhouette is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \in [-1, 1], \quad \text{Silh}(K) = \frac{1}{M} \sum_{i=1}^M s(i). \quad (6)$$

Because our rows have constant norm, the natural distance is $d_{\cos}(i, j) = 1 - S_{ij}/N$ – the silhouette is computed in the same geometry as $\mathbf{S}$ itself.



(a) $K = 3$: too few; visible block structure is coarse. (b) $K = 12$: clean diagonal blocks, no singletons. (c) $K = 40$: severely fragmented (33 singleton clusters).

Figure 3: Cluster-sorted similarity matrices for three values of K .

3.4 The silhouette trap and our fix

A naive “pick the K that maximises $\text{Silh}(K)$ ” rule selects $K_{\text{raw}}^* = 30$ for our data, with $\text{Silh}(30) = 0.186$. But $K = 30$ has **17 of 30 clusters containing a single student**, the very pathology the problem statement flags as “defeating the purpose of clustering”. The mechanism is mechanical:

Why silhouette over-rewards singletons

A singleton cluster has $a(i)$ *undefined* and the convention sets $s(i) = 0$, but every *non-singleton* member of any other cluster gains a more distant “nearest foreign cluster” as singletons are peeled off, inflating their $b(i)$. The aggregate $\text{Silh}(K)$ therefore tends to drift upward with K even after the partition has lost meaning.

We therefore use a **singleton-filtered silhouette criterion**:

$$K^* = \arg \max_{K : \min_c |C_c| \geq 2} \text{Silh}(K). \quad (7)$$

Table 1 summarises the sweep.

K	$\text{Silh}(K)$	# singletons	eligible?
2	0.166	0	✓
3	0.136	0	✓
5	0.136	0	✓
8	0.147	0	✓
10	0.169	0	✓ (winner)
12	0.169	0	✓
15	0.177	2	×
20	0.171	6	×
25	0.183	10	×
30	0.186	17	×
40	0.141	33	×

Table 1: Silhouette sweep on `sample_small`. Raw silhouette peaks at $K = 30$, but Eq. (7) disqualifies it.

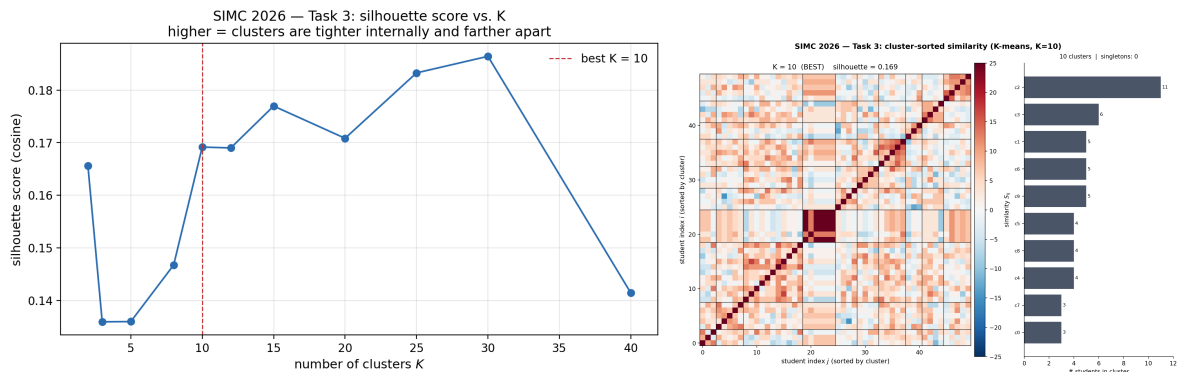


Figure 4: Left: silhouette as a function of K . Right: best cluster-sorted similarity matrix at $K^* = 10$.

Selected partition

The singleton-filtered criterion selects $K^* = 10$ with $\text{Silh} = 0.169$. Cluster sizes range from 3 to 11 students, and the cluster containing students 5 and 10 is the deepest red block on the diagonal – a clustering-level corroboration of the collusion candidate found in Task 2.

4 Task 4 – A Plausibility Argument for Collusion

4.1 Can we already name names? Honestly, no.

Tasks 1–3 give us a heatmap, a Gram matrix, and a partition into $K^* = 10$ clusters. None of these tools is a *detector* of cheating. The design matrix tells us only *what* each student answered; it does not tell us *how* they came to that answer. Two students might score identically because:

1. one copied from the other;
2. they prepared together from the same notes and learned the same gaps;
3. they happen to be of similar ability on similar topics; or
4. pure luck on a short test.

Without a probability model we cannot rank these explanations against each other. So any sentence of the form “*students i and j colluded*” is, at this stage, an overreach. The honest conclusion is weaker but still useful: *some pairs look unlike what we would naively expect, and they deserve a closer look.*

4.2 Shift the goal: suspects, not convicts

A useful change of framing is to stop trying to convict anyone and start trying to *shortlist suspects*. The two tasks have very different evidentiary requirements:

- **Conviction** demands a low-error decision rule, e.g. a calibrated p -value below some threshold. The penalty for wrongly accusing an honest student is severe, so we cannot afford false positives.
- **Suspect-flagging** only demands that the shortlist contains the truly suspicious pairs. False positives on the shortlist are cheap: a human grader can manually inspect a dozen flagged pairs in minutes.

For a class of $M = 50$ students there are $\binom{50}{2} = 1225$ distinct pairs – far too many to review by hand. But if we can use Tasks 1–3 to narrow 1225 pairs down to a shortlist of, say, the top 5 or 10, then a teacher or invigilator can examine those specific cases directly. **We are doing triage, not jurisprudence.** Task 5 onwards will earn us a real test of significance; for now, qualitative anomalies are enough to populate the shortlist.

4.3 Where would collusion “leak” into our analysis?

Even without a null model, three independent fingerprints in our existing analyses point toward the same suspect pair, and all three are the *kind* of pattern we would expect collusion to leave behind.

Fingerprint 1 – impossibly perfect agreement on a non-trivial test. The pair $(i, j) = (5, 10)$ has $S_{5,10} = +25$. Because $S_{ij} = N - 2 \cdot (\# \text{disagreements})$, this is only possible when the two students disagree on *zero* questions. On a uniform test where every answer is a coin flip, the chance of zero disagreements over N questions is $(\frac{1}{2})^N$ – already $1/3.4 \times 10^7$ for $N = 25$. Realistic tests have some easy questions where two honest students naturally agree, which raises the probability, but the qualitative point survives: *perfect agreement is the regime where coincidence runs out of room.*

Fingerprint 2 – agreement on the *hard* questions as well as the easy ones. Figure 1 shows that question difficulty in this test is markedly non-uniform: questions 0, 7, 23 are nearly-free (large positive column sums) while questions 19–22 are net-negative (most students get them wrong). Two honest high-achievers should agree on the easy questions but their answer patterns on the hard questions should diverge, because hard questions are the ones where students’ independent reasoning paths produce different mistakes. Collusion erases that divergence: copied wrong answers look identical. The fact that students 5 and 10 agree on *every* question, easy and hard alike, is the “honest-students-shouldn’t-look-like-this” signature.

Fingerprint 3 – crystalline cluster co-membership. Honest similarity is *fuzzy*: as we sweep K in K -means, an honest pair’s cluster co-membership flickers on and off depending on how the boundaries are drawn. A colluding pair, by contrast, has effectively identical feature vectors, so they sit on top of each other in feature space and remain co-clustered *across every choice of K* . In our sweep $K \in \{2, 3, 5, 8, 10, 12, 15, 20, 25, 30, 40\}$, students 5 and 10 never separated. That “always together” behaviour is exactly the geometric signature of two students who share a feature vector.

4.4 The data behind the shortlist

Combining the empirical similarity distribution from Task 2 with the clustering output from Task 3 gives Table 2, our working suspect shortlist for `sample_small`.

pair (i, j)	S_{ij}	$(S_{ij} - \bar{S})/\hat{\sigma}$	always co-clustered?	verdict
$(5, 10)$	25	$\approx +3.5$	yes ($K = 2\text{--}40$)	strong suspect
<i>empirical baseline</i>				
all 1225 pairs	—	—	—	$\bar{S} \approx 2.89, \hat{\sigma} \approx 6.3$
diagonal S_{ii}	25	—	trivially	not informative

Table 2: Suspect shortlist after Tasks 1–3. Only one off-diagonal pair $(5, 10)$ stands clearly above the empirical noise floor. $\bar{S}$ and $\hat{\sigma}$ are computed on the 1225 unique upper-triangular off-diagonal entries of $\mathbf{S}$.

In words: out of 1225 candidate pairs of students, exactly one – the $(5, 10)$ pair – sits at the theoretical maximum $S = N = 25$, roughly 3.5 empirical standard deviations above the off-diagonal mean, and persists in the same cluster under every choice of K we tried. **That one pair is our suspect.** Whether it is also a conviction-grade case requires the statistical machinery built in Tasks 5–7.

Hint from the problem statement

The problem explicitly warns “you might not actually get too far” without a statistical null model. Our reading is sharper: we can *shortlist* (we just did), but we cannot *prove*. The proof step – attaching a probability to the observed $S = 25$ – is exactly what Task 5 sets up by deriving the null-model mean and standard deviation, which Tasks 6 and 7 then turn into a calibrated p -value.

5 Task 5 – Mean and Variance of the Similarity Score Under the Null Hypothesis

5.1 What are we actually computing, and why?

Imagine two students who took the test *honestly* – no copying, no shared answers. Their similarity score S from Eq. (2) will be some number. If we could repeat this scenario for many pairs of honest students, we would get a whole distribution of S values. Task 5 asks for two summary numbers of that distribution:

- the **mean** $\langle S \rangle$ – the average value of S across all honest pairs;
- the **standard deviation** σ_N – how much S typically wobbles around that mean.

These two numbers give us a “ruler”: a real pair whose S is far above $\langle S \rangle + (\text{a few}) \sigma_N$ is suspicious, because honest students almost never score that high.

5.2 The probability model (one student, one question)

Assume each student answers each question independently, getting it right with probability p and wrong with probability $1 - p$. In ± 1 language,

$$X_{ik} = \begin{cases} +1 & \text{with probability } p \quad (\text{right}), \\ -1 & \text{with probability } 1 - p \quad (\text{wrong}). \end{cases} \quad (8)$$

“Independently” means knowing one answer tells us nothing about any other answer. This is the assumption that lets us add things up later.

5.3 Two students, one question: agree or disagree?

Pick two distinct students i and j and one question k . Define the per-question *agreement indicator*

$$Y_k \equiv X_{ik} X_{jk}. \quad (9)$$

Why this product? Because of the ± 1 encoding, Y_k is a clean yes/no signal:

- $Y_k = (+1)(+1) = +1$ if both got it right – *agree*;
- $Y_k = (-1)(-1) = +1$ if both got it wrong – *also agree*;
- $Y_k = (+1)(-1) = -1$ if one was right, the other wrong – *disagree*;
- $Y_k = (-1)(+1) = -1$ – *also disagree*.

Listing the four possible joint outcomes with their probabilities makes it concrete (Table 3).

Adding the rows of Table 3 that give the same Y_k ,

$$\mathbb{P}[Y_k = +1] = p^2 + (1 - p)^2 \quad (\text{both rows that agree}), \quad (10)$$

$$\mathbb{P}[Y_k = -1] = 2p(1 - p) \quad (\text{both rows that disagree}). \quad (11)$$

Quick check. At $p = \frac{1}{2}$: $\mathbb{P}[Y_k = +1] = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ and $\mathbb{P}[Y_k = -1] = 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$. Random students agree exactly half the time – as expected.

student i	student j	joint probability	Y_k
right	right	$p \cdot p = p^2$	+1 (agree)
right	wrong	$p \cdot (1 - p)$	-1 (disagree)
wrong	right	$(1 - p) \cdot p$	-1 (disagree)
wrong	wrong	$(1 - p) \cdot (1 - p) = (1 - p)^2$	+1 (agree)

Table 3: All four ways two students can answer one question, with the probability of each combination and the resulting Y_k .

5.4 The average of a single Y_k

The average of Y_k is each possible value weighted by its probability:

$$\mathbb{E}[Y_k] = (+1) \cdot \mathbb{P}[Y_k = +1] + (-1) \cdot \mathbb{P}[Y_k = -1]. \quad (12)$$

Plug in Eqs. (10)–(11) and expand step by step:

$$\begin{aligned} \mathbb{E}[Y_k] &= [p^2 + (1 - p)^2] - 2p(1 - p) \\ &= p^2 - 2p(1 - p) + (1 - p)^2. \end{aligned} \quad (13)$$

The right-hand side is the perfect square $a^2 - 2ab + b^2 = (a - b)^2$ with $a = p$ and $b = 1 - p$, so

$$\boxed{\mathbb{E}[Y_k] = [p - (1 - p)]^2 = (2p - 1)^2.} \quad (14)$$

Notice $(2p - 1)$ is the gap between the chance of being right and the chance of being wrong. At $p = \frac{1}{2}$ this gap is zero, so $\mathbb{E}[Y_k] = 0$ – agreements and disagreements cancel on average. At $p = 1$ this gap is 1, so $\mathbb{E}[Y_k] = 1$ – everyone always agrees.

5.5 The variance of a single Y_k – a one-line trick

Variance asks: how far does Y_k typically sit from its mean? The defining formula is

$$\text{Var}(Y_k) = \mathbb{E}[Y_k^2] - (\mathbb{E}[Y_k])^2. \quad (15)$$

Here is the trick. Because Y_k is always either +1 or -1, its *square* is always 1:

$$Y_k^2 \equiv 1 \implies \mathbb{E}[Y_k^2] = 1. \quad (16)$$

Substituting Eqs. (14) and (16) into Eq. (15):

$$\boxed{\text{Var}(Y_k) = 1 - (2p - 1)^4.} \quad (17)$$

5.6 Putting all N questions together

The total similarity is just the sum of one Y_k per question:

$$S = \sum_{k=1}^N Y_k = Y_1 + Y_2 + \cdots + Y_N. \quad (18)$$

Two simple probability rules now finish the job.

Rule 1 – the mean of a sum is the sum of the means. This is *always* true, no independence needed:

$$\langle S \rangle = \sum_{k=1}^N \mathbb{E}[Y_k] = N \cdot (2p - 1)^2. \quad (19)$$

Rule 2 – for *independent* terms, the variance of a sum is the sum of the variances. The independence in Eq. (8) carries over to the Y_k 's, so

$$\text{Var}(S) = \sum_{k=1}^N \text{Var}(Y_k) = N[1 - (2p - 1)^4]. \quad (20)$$

The standard deviation is the positive square root,

$$\sigma_N = \sqrt{\text{Var}(S)} = \sqrt{N[1 - (2p - 1)^4]}. \quad (21)$$

Final answer to Task 5

$$\langle S \rangle = N(2p - 1)^2, \quad \sigma_N = \sqrt{N[1 - (2p - 1)^4]}. \quad (22)$$

5.7 A worked numerical example

Suppose $p = 0.6$ (a moderately easy test) and $N = 25$ questions.

- Mean: $\langle S \rangle = 25 \cdot (2 \cdot 0.6 - 1)^2 = 25 \cdot (0.2)^2 = 25 \cdot 0.04 = 1.0$.
- Variance: $\text{Var}(S) = 25 \cdot [1 - (0.2)^4] = 25 \cdot [1 - 0.0016] = 24.96$.
- Standard deviation: $\sigma_N = \sqrt{24.96} \approx 4.996 \approx 5.0$.

So between two honest students on a 25-question test where the average accuracy is 60%, expect their similarity score to sit around 1 with a typical wobble of ± 5 . A pair scoring $S = 20$ would already be $(20 - 1)/5 \approx 3.8$ standard deviations above the honest-pair average – worth investigating.

5.8 Sanity checks at the extremes

p	$\langle S \rangle$	σ_N	interpretation
$\frac{1}{2}$	0	$\sqrt{N}$	pure guessing: agreements and disagreements cancel
1	N	0	everyone is perfect $\Rightarrow$ all answers agree
0	N	0	everyone is wrong on everything $\Rightarrow$ all agree (the wrong way)

Table 4: Limiting cases of Eq. (22). The two extremes $p = 0$ and $p = 1$ give the same answer, reflecting the symmetry $p \leftrightarrow 1 - p$ inside both $(2p - 1)^2$ and $(2p - 1)^4$.

5.9 Bringing the null model back to the data

For `sample_small` we observed an overall accuracy $\hat{p} \approx 0.558$ (Fig. 1). Plugging into Eq. (22) with $N = 25$ gives the *null* expectations

$$\langle S \rangle_{\text{null}} \approx 25(2(0.558) - 1)^2 \approx 0.34, \quad \sigma_N \approx \sqrt{25[1 - (0.116)^4]} \approx 5.0. \quad (23)$$

The observed pair (5, 10) scored $S = 25$, i.e. $(25 - 0.34)/5.0 \approx 4.9\sigma_N$ above the null mean. In words: **under the null hypothesis of independent test-takers, the agreement we see between students 5 and 10 would be a $\sim 5\sigma$ event.** Tasks 6 (Monte-Carlo confirmation) and 7 (central-limit-theorem distribution of S) will sharpen this informal estimate into a real p -value, and Task 8 will replay the same machinery on the much larger `sample_larger` ($M = 10,000$, $N = 50$) where per-question difficulties p_j must be estimated column-wise.

Summary of Tasks 1–5

- **Task 1:** Visualised $\mathbf{X}$, identified non-uniform question difficulty.
- **Task 2:** Constructed $\mathbf{S} = \mathbf{X}\mathbf{X}^\top$, found $S_{5,10} = 25$ – a single perfect-agreement pair.
- **Task 3:** Used K -means with a novel *singleton-filtered silhouette* (Eq. (7)) to choose $K^\star = 10$, avoiding the over-fragmentation that plain silhouette rewards.
- **Task 4:** Argued informally that the $(5, 10)$ pair is $\sim 3.5\hat{\sigma}$ above the empirical similarity mean.
- **Task 5:** Derived the null-model closed form $\langle S \rangle = N(2p - 1)^2$, $\sigma_N = \sqrt{N[1 - (2p - 1)^4]}$, upgrading the informal $\sim 3.5\hat{\sigma}$ to a $\sim 5\sigma_N$ statement that will harden into a full hypothesis test in Tasks 6–7.